

Homework 6

BSTA 551

Directions

For simulation problems, set your seed to `set.seed(551)` for reproducibility.

For the qmd of this Homework page, see this [github link](#)

You must show all of your work to receive credit.

This homework covers material from Lessons 15, 16, and 17: Neyman-Pearson Lemma and Likelihood Ratio Tests; Further Aspects of Hypothesis Testing and CI-Test Duality; Bootstrap Hypothesis Testing.

Questions

Problem 1: Neyman-Pearson Lemma (Lesson 15)

Adverse events in a clinical trial are monitored over successive reporting windows. The number of adverse events per window follows $X_i \stackrel{iid}{\sim} \text{Poisson}(\lambda)$. A safety board tests whether the adverse event rate has doubled from the historical baseline.

$$H_0 : \lambda = 2 \quad \text{vs} \quad H_a : \lambda = 4$$

based on $n = 6$ reporting windows.

(a) Compute the Neyman-Pearson likelihood ratio $\Lambda^* = L(\lambda = 4)/L(\lambda = 2)$ and show that the NP rejection region $\Lambda^* \geq k$ simplifies to $\sum_{i=1}^6 X_i \geq c$ for some constant c . (*Hint: the joint PMF is $e^{-n\lambda} \lambda^{\sum x_i} / \prod x_i!$*)

(b) Under H_0 , $Y = \sum_{i=1}^6 X_i \sim \text{Poisson}(12)$. Find the critical value c such that the Type I error $\alpha^* = P_{H_0}(Y \geq c) \leq 0.05$. Use `ppois()` in R. Report the exact achieved α^* .

(c) Compute the power of this test at $\lambda = 4$ (i.e., $1 - \beta^*$). Under H_a , $Y \sim \text{Poisson}(24)$.

Problem 2: Likelihood Ratio Test and Wilks' Theorem (Lesson 15)

In a study of hospital-acquired infections, the number of infections per 100 patient-days follows $X_i \stackrel{iid}{\sim} \text{Poisson}(\lambda)$. The national benchmark rate is $\lambda_0 = 3.5$ infections per 100 patient-days. A hospital records the following counts over $n = 10$ monthly reporting periods:

$$X = \{5, 2, 4, 6, 3, 5, 4, 7, 3, 4\}$$

Test $H_0 : \lambda = 3.5$ vs $H_a : \lambda \neq 3.5$.

(a) Write down the log-likelihood $\ell(\lambda) = \sum_{i=1}^n \ln f(x_i; \lambda)$ for the Poisson model and find the MLE $\hat{\lambda}$.

(b) The LRT statistic is $\Lambda = L(\lambda_0)/L(\hat{\lambda})$. Compute $-2\ln(\Lambda)$ using the formula:

$$-2\ln(\Lambda) = -2 \left[n\hat{\lambda} - n\lambda_0 + n\bar{x} \ln\left(\frac{\lambda_0}{\hat{\lambda}}\right) \right]$$

using the observed data. (*Show the numerical calculation.*)

(c) By Wilks' theorem, $-2\ln(\Lambda) \xrightarrow{d} \chi_1^2$ under H_0 . Compute the p-value and state your conclusion at $\alpha = 0.05$. Also compute the exact p-value using `poisson.test()` and compare.

(d) How many free parameters does H_0 constrain? Why does the χ^2 approximation have 1 degree of freedom here?

Problem 3: Statistical vs. Practical Significance (Lesson 16)

A large randomized controlled trial ($n = 3,200$ patients per arm) evaluates a new lipid-lowering drug. The primary outcome is LDL cholesterol (mg/dL) measured after 6 months. Results:

- Control arm: $\bar{x}_{\text{control}} = 142.1$ mg/dL
- Treatment arm: $\bar{x}_{\text{treat}} = 140.3$ mg/dL

Assume equal and known $\sigma = 28.15$ mg/dL and use a two-sample z-test.

- (a) Compute the observed z-statistic, two-sided p-value, and 95% CI for $\mu_{\text{treat}} - \mu_{\text{control}}$.
 - (b) Compute Cohen's $d = |\bar{x}_{\text{treat}} - \bar{x}_{\text{control}}|/\sigma$. Clinical guidelines require a reduction of ≥ 10 mg/dL to be meaningful. Does the CI provide evidence of a clinically meaningful reduction?
 - (c) In 2–3 sentences, explain how a study can produce a statistically significant result that is not clinically significant, and why the CI is more informative than the p-value alone.
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Problem 4: CI-Test Duality — Theory and Computation (Lesson 16)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poisson}(\lambda)$. The exact two-sided $100(1 - \alpha)\%$ confidence interval for λ based on $T = \sum_{i=1}^n X_i$ is:

$$\left(\frac{\chi_{\alpha/2, 2T}^2}{2n}, \frac{\chi_{1-\alpha/2, 2T+2}^2}{2n} \right)$$

- (a) State the CI-Test Duality Theorem precisely. Then argue that the Poisson CI above is dual to the exact level- α test of $H_0 : \lambda = \lambda_0$ vs $H_a : \lambda \neq \lambda_0$.
 - (b) Using the hospital infection data from Problem 2 ($n = 10$, $\sum x_i = 43$):
 - (i) Compute the 95% Poisson CI using `qchisq()`.
 - (ii) Test $H_0 : \lambda = 3.5$ using `poisson.test()`. Is the test decision consistent with CI membership?
 - (iii) Test $H_0 : \lambda = 4.5$. Does $\lambda_0 = 4.5$ fall inside or outside the CI? Is this consistent?
 - (c) Compute the 95% lower confidence bound for λ and use it to test $H_0 : \lambda \leq 3.5$ vs $H_a : \lambda > 3.5$ at $\alpha = 0.05$.
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Problem 5: Constructing CIs and the p-Value Function (Lesson 16)

The plasma concentration (mg/L) of a drug is measured for $n = 12$ patients:

$$X = \{1.8, 2.6, 3.1, 2.2, 4.7, 2.9, 3.8, 2.4, 1.6, 3.5, 4.1, 2.7\}$$

The regulatory target mean concentration is $\mu_0 = 3.0$ mg/L.

(a) Compute the 95% two-sided t-interval for μ . Without computing a test statistic, use the duality theorem to determine whether you would reject $H_0 : \mu = 3.0$ at $\alpha = 0.05$.

(b) Starting from the pivot $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$, compute a 95% two-sided CI for σ^2 and for σ using the sample data. Use `qchisq()`.

(c) Sweep μ_0 over $[1.5, 4.5]$ in steps of 0.05. For each value compute the two-sided t-test p-value. Plot the p-value function with the 95% CI shaded as the region where $p(\mu_0) \geq 0.05$. Verify visually that the CI from part (a) corresponds exactly to this region.

```
mu0_grid <- seq(1.5, 4.5, by = 0.05)
```

Problem 6: Bootstrap Hypothesis Testing (Lesson 17)

Serum ferritin (ng/mL) is a biomarker for iron stores. A study of $n = 18$ patients with chronic kidney disease gives:

```
ferritin <- c(28.4, 45.2, 19.8, 67.3, 31.5, 88.1, 22.6, 54.9,  
             41.2, 76.8, 33.7, 29.4, 58.3, 44.1, 92.5, 37.6,  
             25.8, 61.4)
```

The population norm for healthy adults is $\mu_0 = 40$ ng/mL. Test $H_0 : \mu = 40$ vs $H_a : \mu \neq 40$ at $\alpha = 0.05$.

(a) Make a histogram of the data. Does normality seem reasonable? Compute $T_{\text{obs}} = (\bar{x} - \mu_0)/(s/\sqrt{n})$ and the parametric t-test p-value.

(b) Implement the bootstrap test with $B = 5,000$ resamples (`set.seed(551)`): shift the data so $\bar{x}^* = \mu_0$, resample with replacement, compute bootstrap t-statistics, and compute the p-value $\hat{p} = B^{-1} \sum \mathbf{1}(|T^{*b}| \geq |T_{\text{obs}}|)$. Compare to the t-test p-value.